CLAIMS

What is claimed is:

1. A method for negotiating a data encoding scheme between two devices connected by a parallel communication interface, the method comprising the steps of:

    - establishing a communication session using a default, pre-defined encoding map;
    - transmitting, from a first device to a second device, a capabilities message listing one or more available high-density encoding maps;
    - transmitting, from the second device to the first device, a selection message indicating a chosen encoding map from the list; and
    - conducting subsequent data transmission for the session using the chosen high-density encoding map.

2. The method of claim 1, wherein the negotiation occurs at the Physical Layer (Layer 1) or Data-Link Layer (Layer 2) of the OSI model.

3. The method of claim 1, wherein the default encoding map is a low-cardinality map and the chosen encoding map is a high-cardinality map.

4. The method of claim 1, wherein the parallel communication interface comprises a two-dimensional array of pixels.

5. The method of claim 4, wherein the default encoding map represents a binary state using black and white pixel values.

6. The method of claim 4, wherein the chosen high-density encoding map maps multi-byte data blocks to unique pixel color values.

7. The method of claim 1, wherein the parallel communication interface is a massively parallel optical interface as described in U.S. Patent Application No. 19/559,705.